

Network Monitoring & Security Operations Lab

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Project Type: Cybersecurity & Network Security Lab

Platform: Cisco Packet Tracer

Overview

This project demonstrates an enterprise-style network security and monitoring environment built in Cisco Packet Tracer. It covers VLAN segmentation, inter-VLAN routing, SSH remote management, switch port security, unused-port hardening, ACL enforcement, centralized Syslog monitoring, and security verification.

Objectives

- Multi-VLAN enterprise-style network
- Inter-VLAN routing
- Management VLAN
- Port security and unused-port hardening
- SSH remote management
- Extended ACL enforcement
- Centralized Syslog
- Connectivity and security verification

Network Devices

Device	Type	Role
R1-EDGE	Cisco 2911	Simulated edge/upstream router
R2-CORE	Cisco 2911	Inter-VLAN routing, ACL and Syslog
SW1-HQ	Cisco 2960	Core/distribution switching
SW2-HQ	Cisco 2960	HR/Finance access
SW3-ACCESS	Cisco 2960	Sales/IT access
SW4-ACCESS	Cisco 2960	Server/Management access
SRV1-MONITORING	Server-PT	Central Syslog server
PC1-PC10	PC-PT	HR, Finance, Sales, IT and Admin endpoints

VLAN and IP Addressing Plan

VLAN	Department	Network	Gateway
10	HR	10.10.10.0/24	10.10.10.1
20	Finance	10.10.20.0/24	10.10.20.1
30	Sales	10.10.30.0/24	10.10.30.1
40	IT	10.10.40.0/24	10.10.40.1
50	Servers	10.10.50.0/24	10.10.50.1
99	Management	10.10.99.0/24	10.10.99.1

Security Controls

VLAN Segmentation

Separate VLANs are used for HR, Finance, Sales, IT, Servers, and Management.

Inter-VLAN Routing

R2-CORE provides the default gateway for each VLAN through G0/0 subinterfaces.

SSH Remote Management

R2 and all switches use SSH version 2 with local authentication. Credentials are excluded.

Port Security

Active access ports use one sticky MAC address with violation mode restrict.

Unused-Port Hardening

Unused switch interfaces were administratively shut down.

ACL Enforcement

HR-RESTRICTION denies HR-to-Finance traffic while permitting HR traffic to other destinations.

Centralized Syslog

R2 and switches send logs to SRV1-MONITORING at 10.10.50.10 using UDP 514.

Verification Commands

```
show ip interface brief
show vlan brief
show interfaces trunk
show ip access-lists HR-RESTRICTION
show logging
show ip ssh
show port-security interface
```

Evidence Screenshots

1. 01-enterprise-topology.png
2. 02-management-vlan-verification.png
3. 03-sw2-management-verification.png
4. 04-sw3-management-verification.png
5. 05-sw4-management-verification.png
6. 06-ssh-remote-management.png
7. 07-port-security-baseline.png
8. 08-acl-enforcement.png
9. 09-acl-policy-verification.png
10. 10-syslog-r2-verification.png
11. 11-central-syslog-verification.png
12. 12-syslog-server-configuration.png
13. 13-final-r2-security-verification.png

Skills Demonstrated

Cisco Packet Tracer; Cisco IOS; TCP/IP; IPv4; VLANs; trunking; inter-VLAN routing; extended ACLs; SSH; port security; network hardening; Syslog monitoring; troubleshooting; security verification; technical documentation.